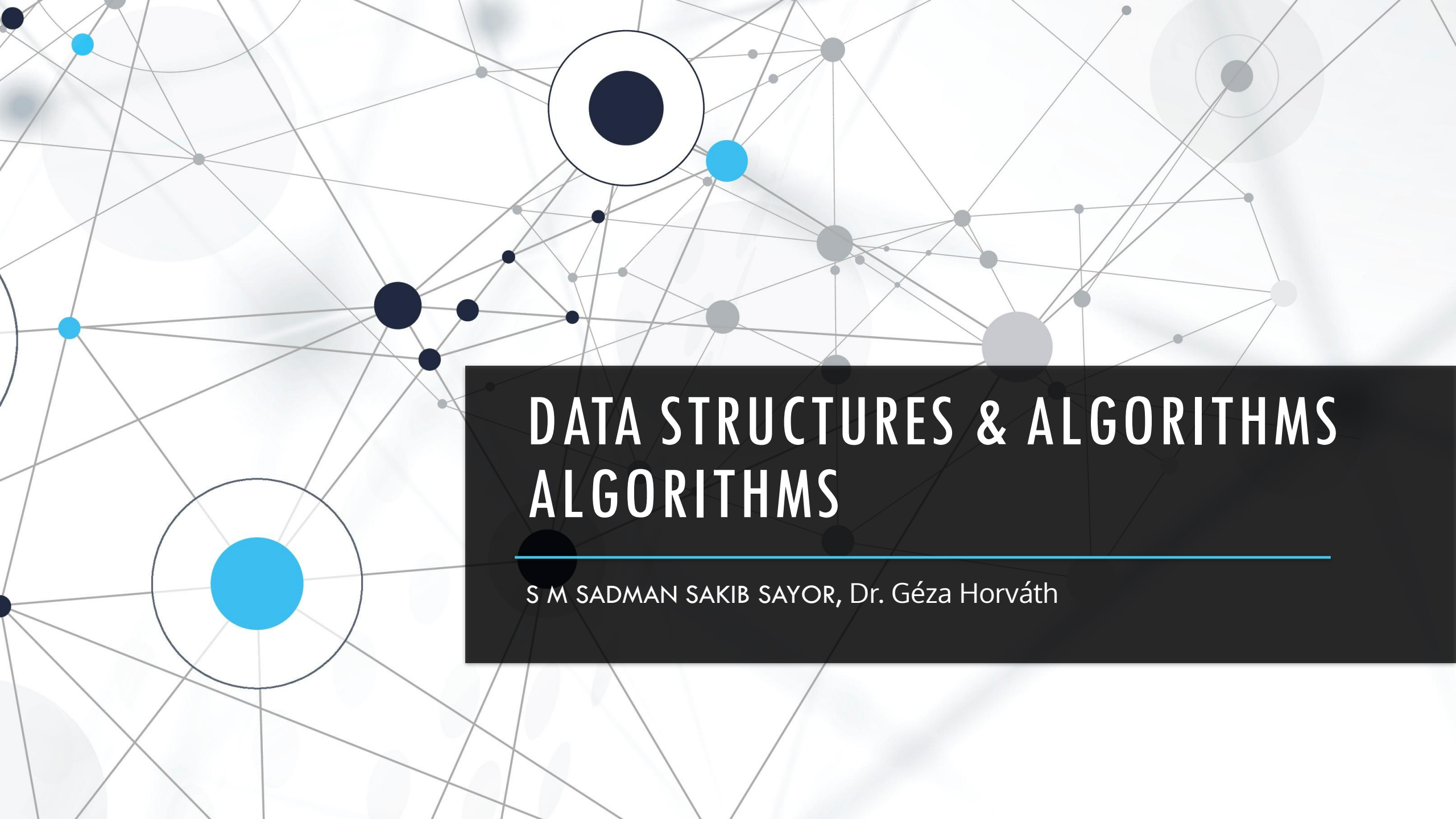


The background features a complex network diagram with numerous nodes of varying sizes (dark blue, light blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger concentric circles. The overall aesthetic is modern and technical.

DATA STRUCTURES & ALGORITHMS ALGORITHMS

S M SADMAN SAKIB SAYOR, Dr. Géza Horváth

SEMESTER CONTENT

First Half

- ☐ Introduction to DSA
- ☐ Design and Analysis of Algorithm
- ☐ Asymptotic Notation
- ☐ Set, Multi-set, Array and Matrix
- ☐ Stack, Queue and List
- ☐ Table and Hash-table

Second Half

- ☐ Tree , Binary-tree and BST
- ☐ AVL Tree
- ☐ Red-black Trees
- ☐ Graphs
- ☐ Parallel Algorithm
- ☐ Discussion for Exam

CONCEPT QUESTIONS - 1

If I ask you to find a math book from
the **STACK**,
How will you find it?



CONCEPT QUESTIONS - 2

If I ask you to find a math book from
the **BOOK SHELF**,
How will you find it?



CONCEPT QUESTIONS - 3

If I ask you to find a math book
from the **LIBRARY**,
How will you find it?



TODAY'S TOPICS

- ❑ Data
- ❑ Data Structures
- ❑ Algorithms
- ❑ Algorithms Analysis
- ❑ Classification of Data Structures
- ❑ Application of DSA

WHAT IS DATA?

Data is a set of discrete or continuous values that convey information by representing quantity, quality, fact, statistics, or other basic units of meaning, or simply sequences of symbols that can be formally understood.

It is commonly arranged into tables to provide more context and definition to it.



EXAMPLES OF DATA

The term "data" was first used to refer to "transmissible and storable computer information" in 1946.

Examples of Data:

- name
- height
- address
- temperature
- salary
- device model number

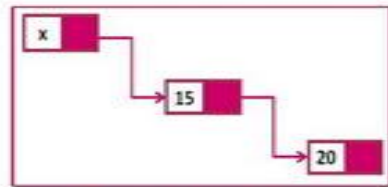


WHAT IS DATA STRUCTURE?

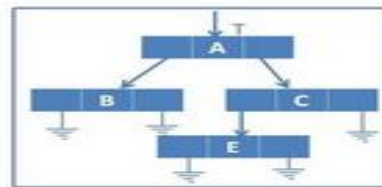
Data Structure is a branch of computer Science that deals with how data can be stored, organized, and retrieved in the most efficient way possible



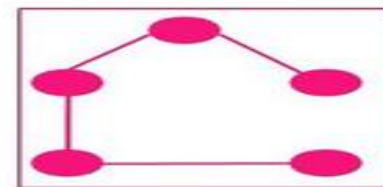
Sorting



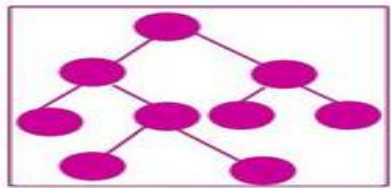
Link list



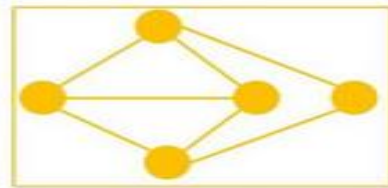
list



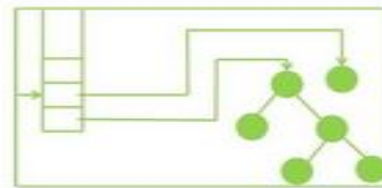
spanning tree



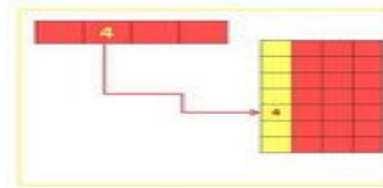
Tree



Graph



Stack

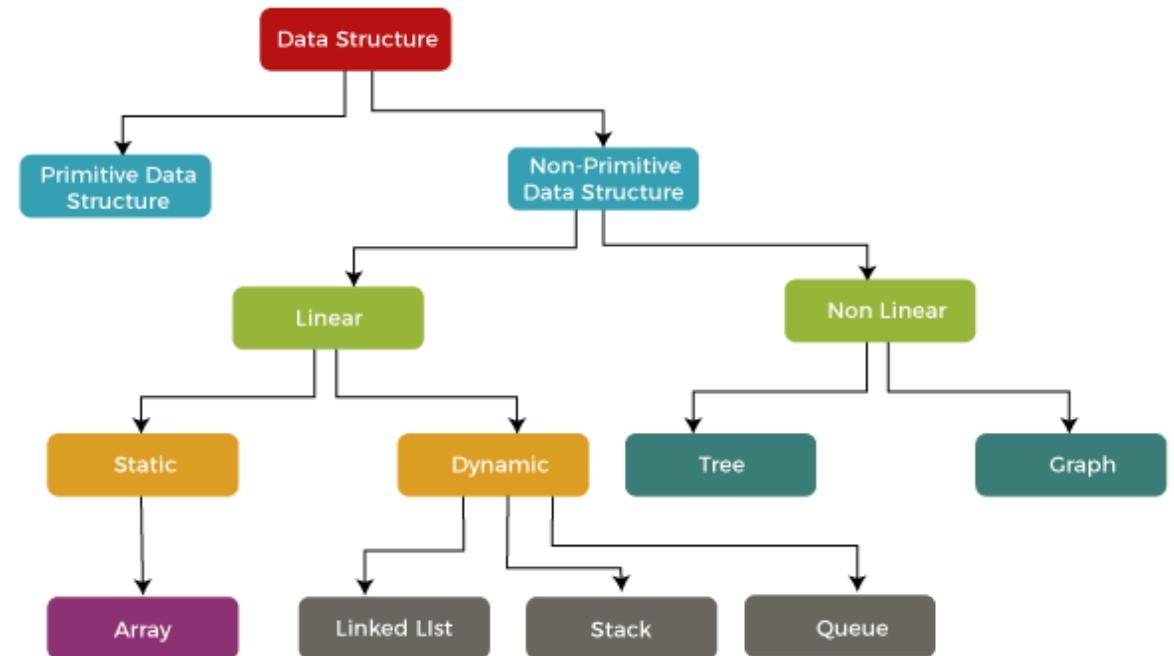


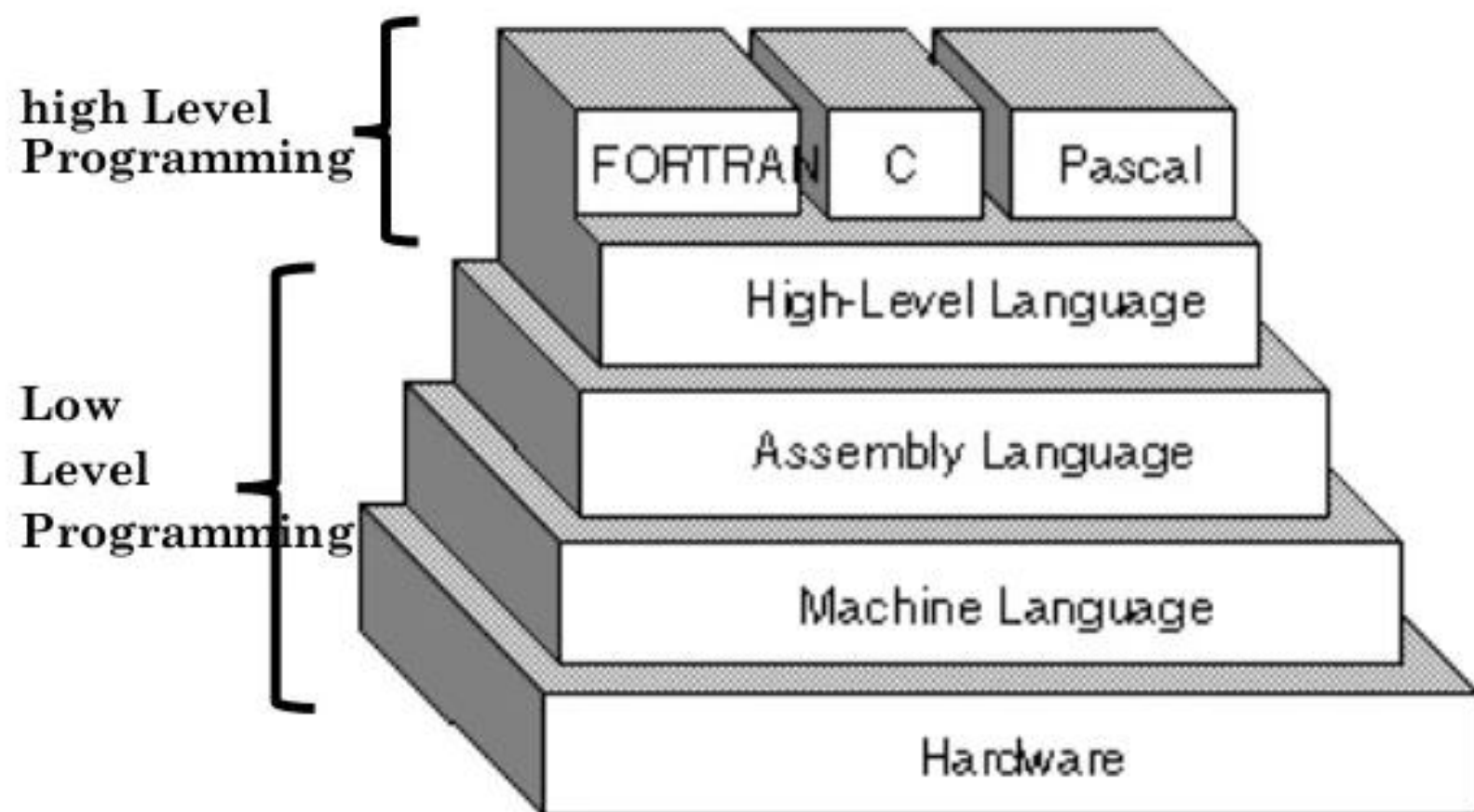
Hashing

CLASSIFICATION OF DATA STRUCTURE

Data Structures can be classified into two categories:

- Primitive Data Structure
- Non-Primitive Data Structure





WHY LEARN DATA STRUCTURE?

Two of the most significant areas of computer science are data structure and algorithms.

- Proficiency in data structures enable us to organize and store data in the most efficient way possible.
- Proficiency in algorithms enable us to process that data more meaningfully.

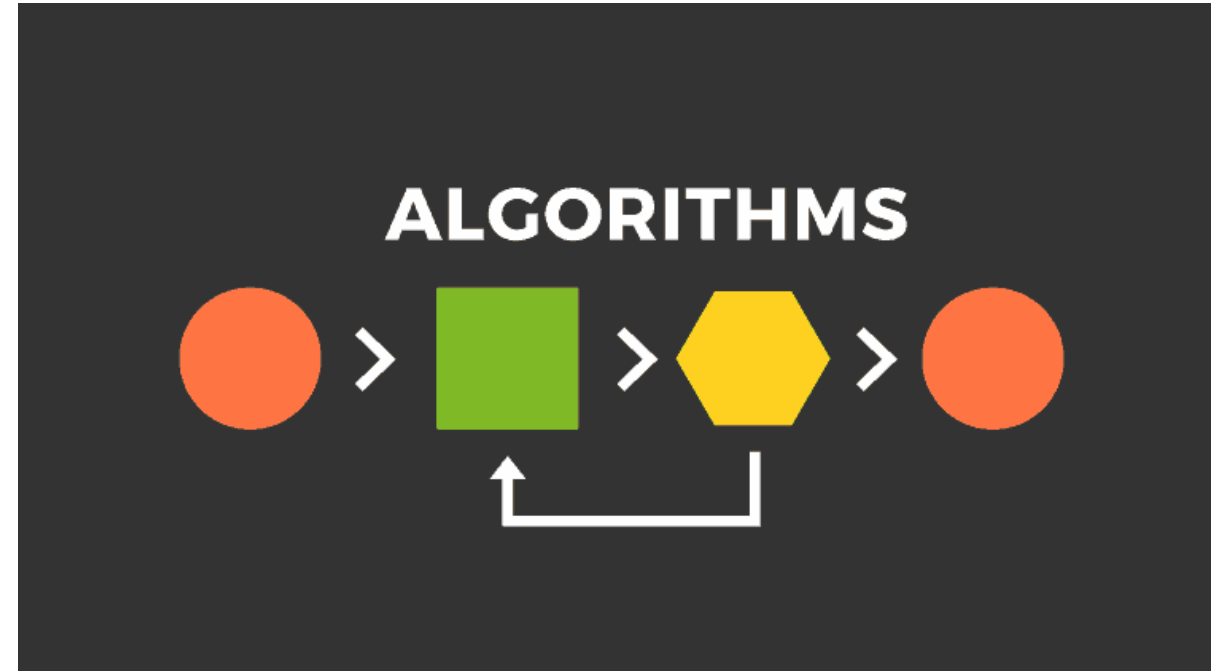


ALGORITHM

Lets simplify our earlier definations...

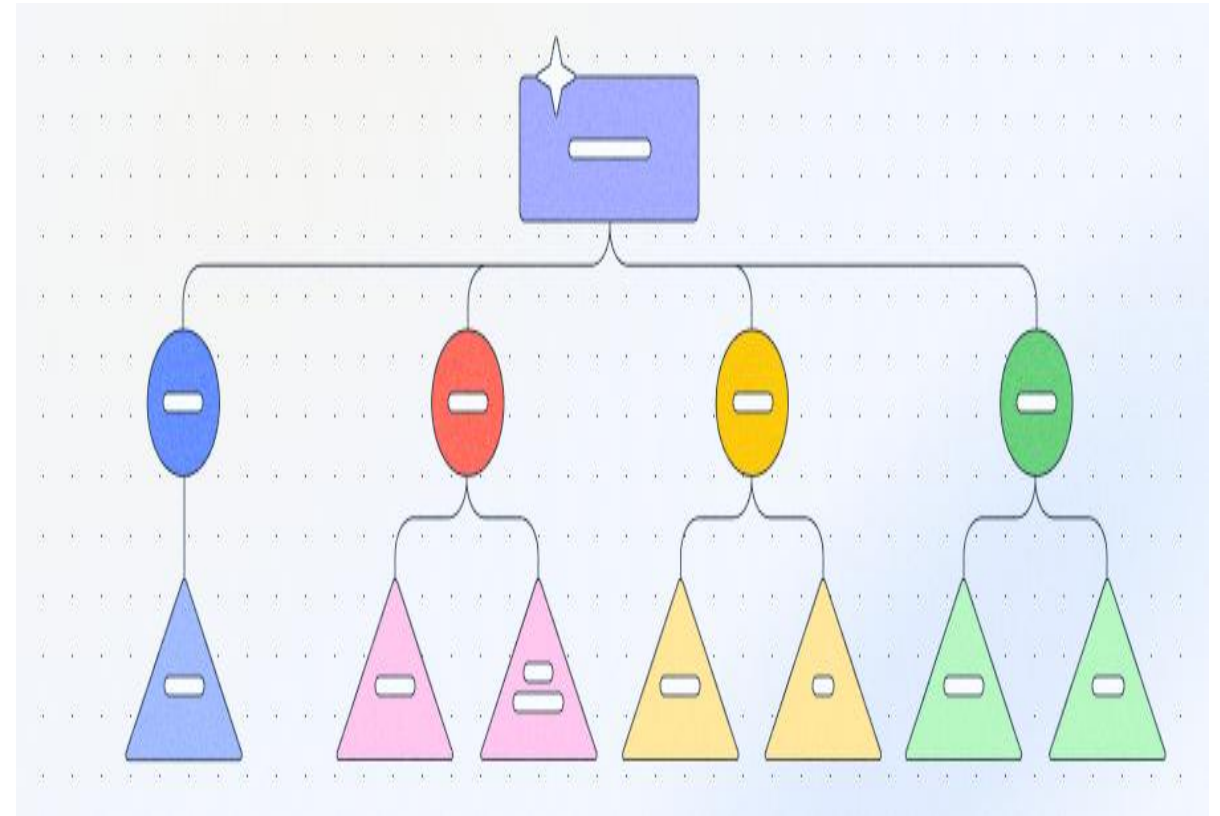
A **data structure** is a way to organize information.

A **algorithm** is a way to process information to reach and end goal.



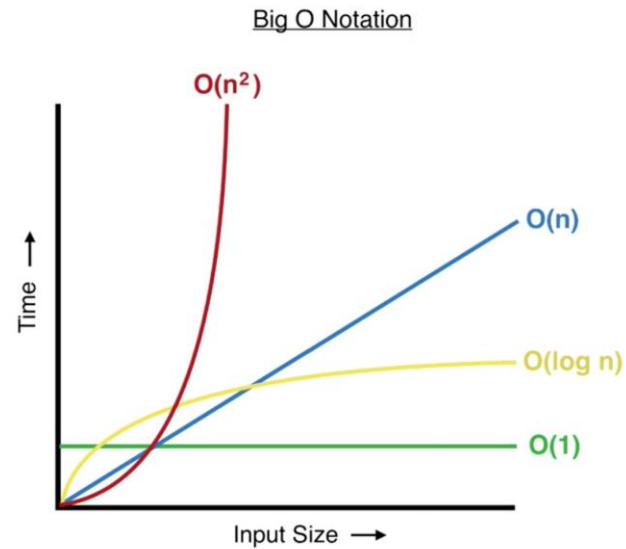
EXAMPLES OF ALGORITHMS

- Sorting
- Binary Search
- Linear Search
- Bit Manipulation
- In-order
- Pre-order
- Post-order
- DFS/BFS
- Topological Sort
- DIJKSTRA
- Bellman Ford
- Greedy Algorithm
- Divide & Conquer
- Backtracking

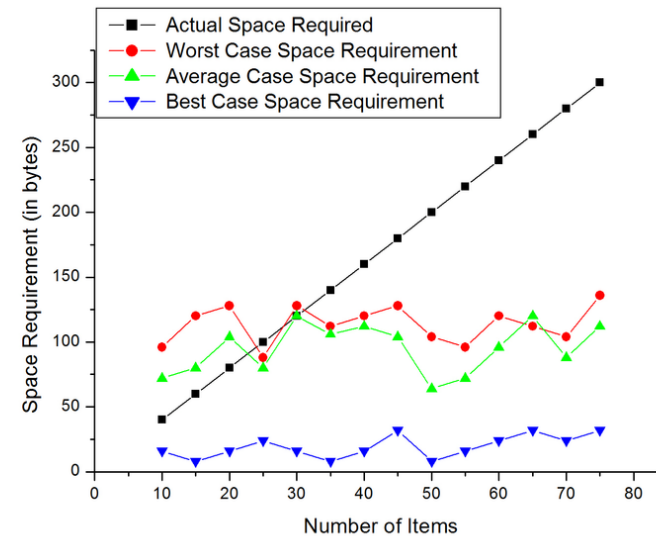


ALGORITHMS ANALYSIS

Time Complexity

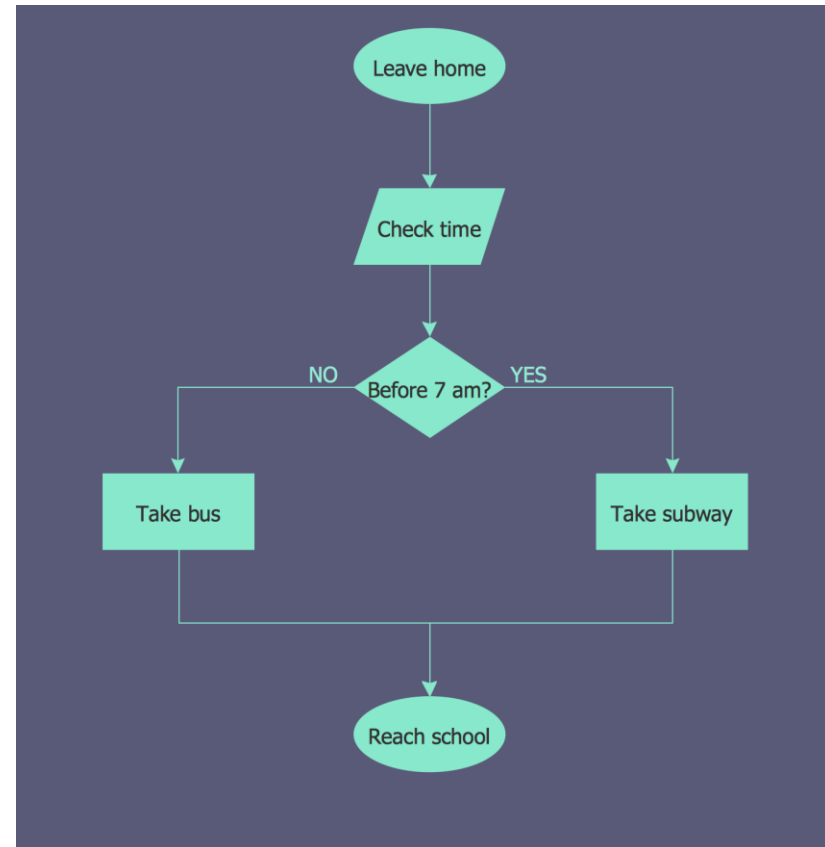


Space Complexity



FLOWCHART

A **flowchart** is a visual representation of the steps and decisions involved in an algorithm.



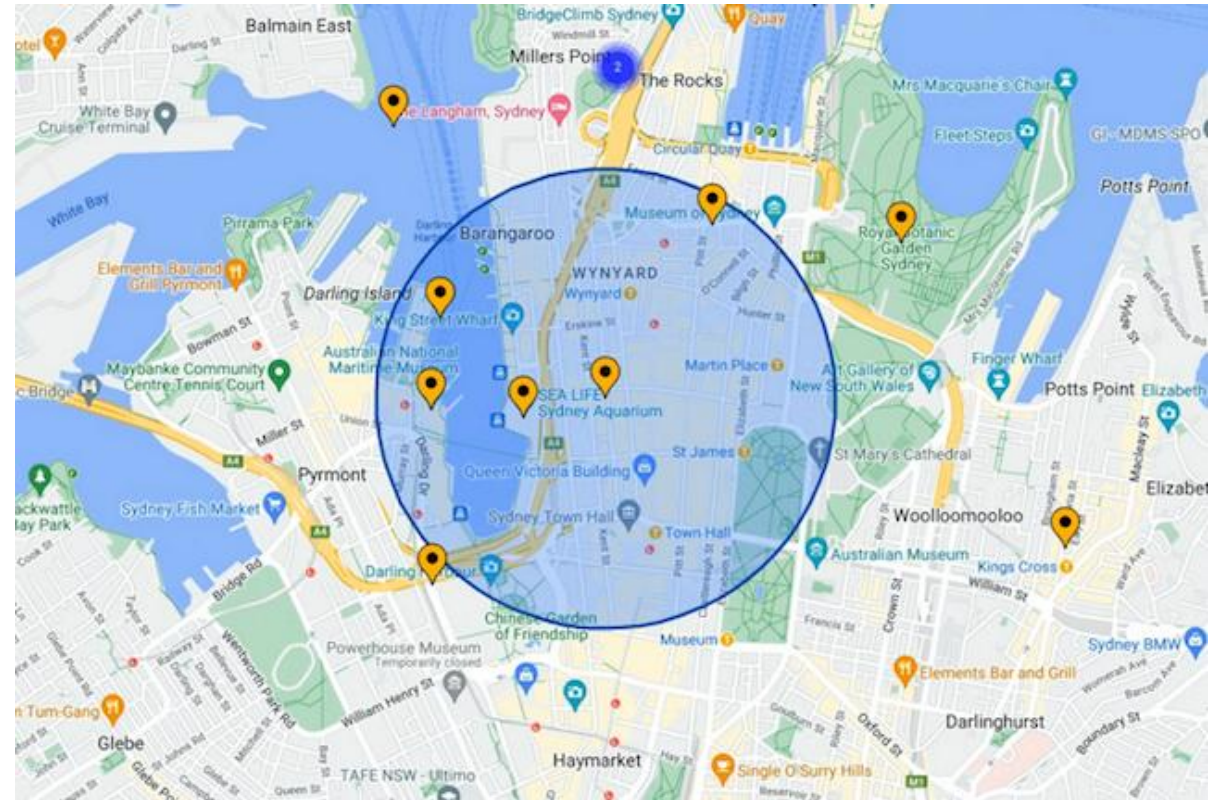
CONCEPT QUESTIONS - 4

Why the contacts in phone are saved/shown in alphabetical order?



APPLICATION OF DSA

Google Maps uses a **graph** data structure and employs **Dijkstra's algorithm** to find the shortest path between two locations.



POINT TO PONDER



CLASS ACTIVITY- 1

Rules

- The top 6 students out of 41 will receive exclusive prizes.
- The quiz is time-limited to a maximum of 12 minutes.
- Each quiz has a total score of 10 points.
- The use of any electronic devices is strictly prohibited during the quiz, and no external resources may be used.

